
Electric Circuits

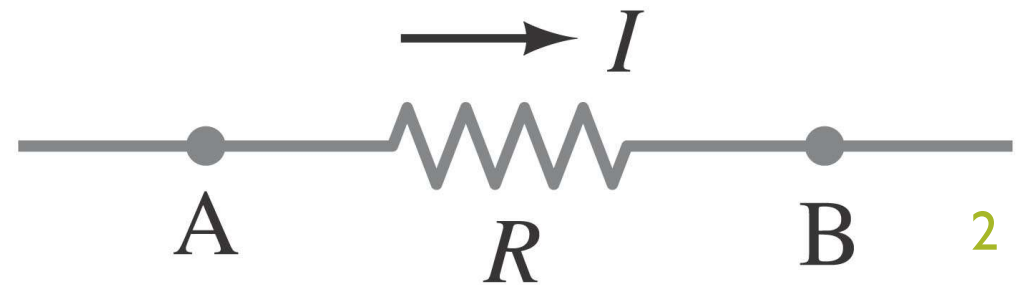
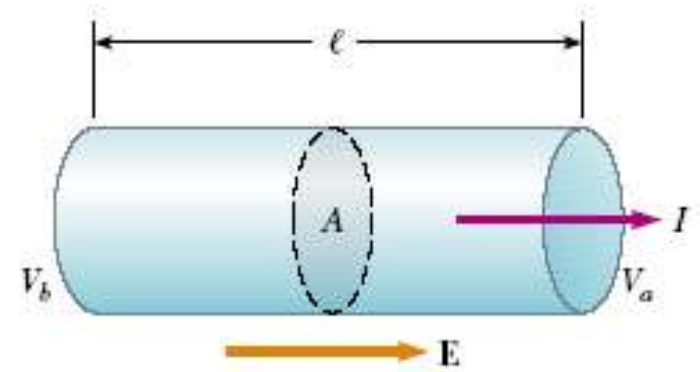
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Ohm's Law

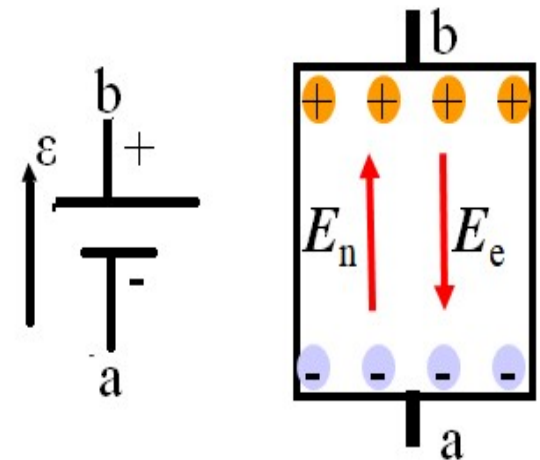
- The quantity $(l / \sigma A)$ is called the resistance R of the conductor. It is defined as the ratio of the potential difference across a conductor to the current through the conductor:

$$R = \frac{\Delta V}{I} = \frac{l}{\sigma A} = \frac{\rho l}{A}$$



Electromotive Force (emf)

- The electromotive force (emf), ε , of a battery is the maximum possible voltage that the battery can provide between its terminals.
 - *The emf supplies energy, it does not apply a force.*
- The battery will normally be the source of energy in the circuit.
- The positive terminal of the battery is at a higher potential than the negative terminal.



Power

- The total power output of the battery is

$$P = I \Delta V = I \varepsilon$$

- The delivered power to the external resistor is

$$P = I^2 R$$

Kirchhoff's Rules



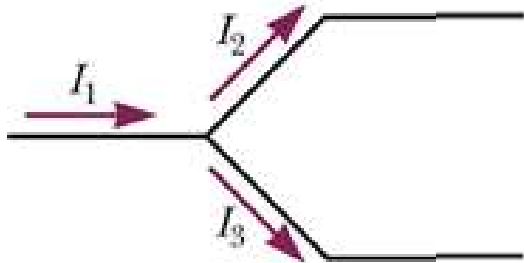
Kirchhoff's Current Law (KCL)

- It is applied at nodes (junctions with more than two branches)
- It expresses the concept of *Conservation of Charges*
- The sum of the currents at any junction must equal zero
- In other words, the sum of the input currents to the node equals to the sum of the output currents from that node.

$$\sum I_{in} = \sum I_{out}$$

Kirchhoff's Current Law (KCL)

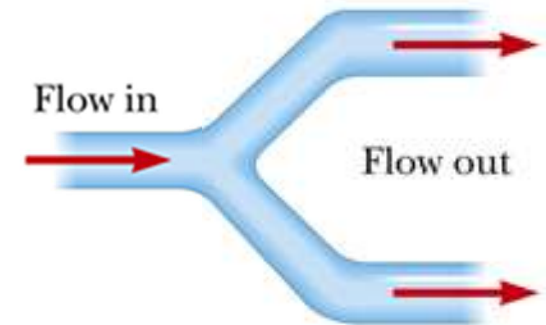
The amount of charge flowing out of the branches on the right must equal the amount flowing into the single branch on the left.



$$I_1 = I_2 + I_3$$

$$\sum I_{in} = \sum I_{out}$$

The amount of water flowing out of the branches on the right must equal the amount flowing into the single branch on the left.



Kirchhoff's Voltage Law (KVL)

- It is applied at Loops (closed paths)
- It expresses the concept of *Conservation of Energy*
- The sum of the potential differences across all elements around any closed circuit loop must be zero.
- Then, in a certain direction, for any loop;

$$\sum \varepsilon = \sum IR$$

Solving DC Circuits, Using Kirchhoff's Laws

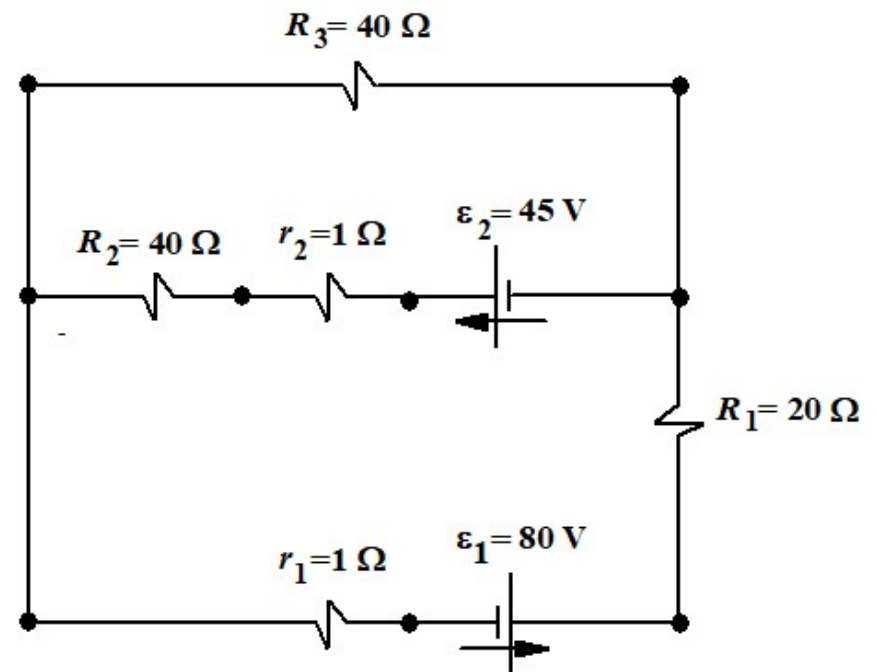
- Assign currents in All Branches in the circuit
- The direction is arbitrary, but you must adhere to the assigned directions when applying Kirchhoff's rules.
- Apply KCL to any node in the circuit that provides new relationships among the various currents.
- Apply KVL to as many loops as are needed to solve for the unknowns.
 - To apply the loop rule, you must choose a direction in which to travel around the loop.
 - You must also correctly identify the potential difference as you cross various elements.

Solving DC Circuits, Using Kirchhoff's Laws

- Solve the equations simultaneously for the unknown quantities.
- Check your numerical answers for consistency.
 - If any current value is negative, it means that the assumed direction of that current incorrectly. *It must be reversed*
 - The magnitude will still be correct.

Example I:

- In the shown circuit, determine the currents in all resistors by applying Kirchhoff's Rules



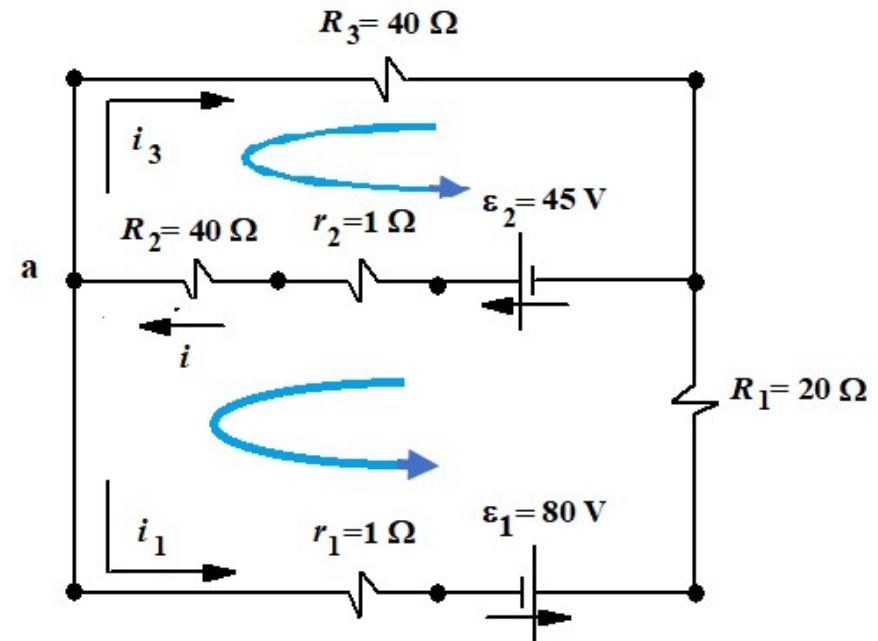
Solution:

- i_1 & i_2 and i_3 are assumed in the branches of the circuit as shown
- Applying KCL at node 'a',

$$i_2 = i_3 + i_1$$

- Applying KVL at the lower Loop,

$$i_1(r_1 + R_1) + i_2(r_2 + R_2) = \mathcal{E}_1 + \mathcal{E}_2 \Rightarrow 21i_1 + 41i_2 = 125$$



Solution (Continue)

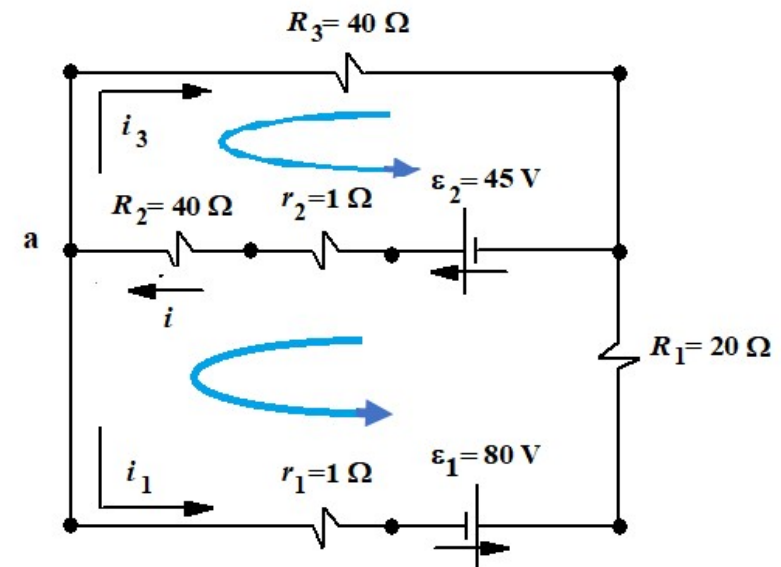
- Applying KVL at the upper loop,

$$i_2(r_2 + R_2) + i_3 R_3 = \mathcal{E}_2 \quad \Rightarrow \quad 41 i_2 + 40 i_3 = 45$$

- Solving the three equations,

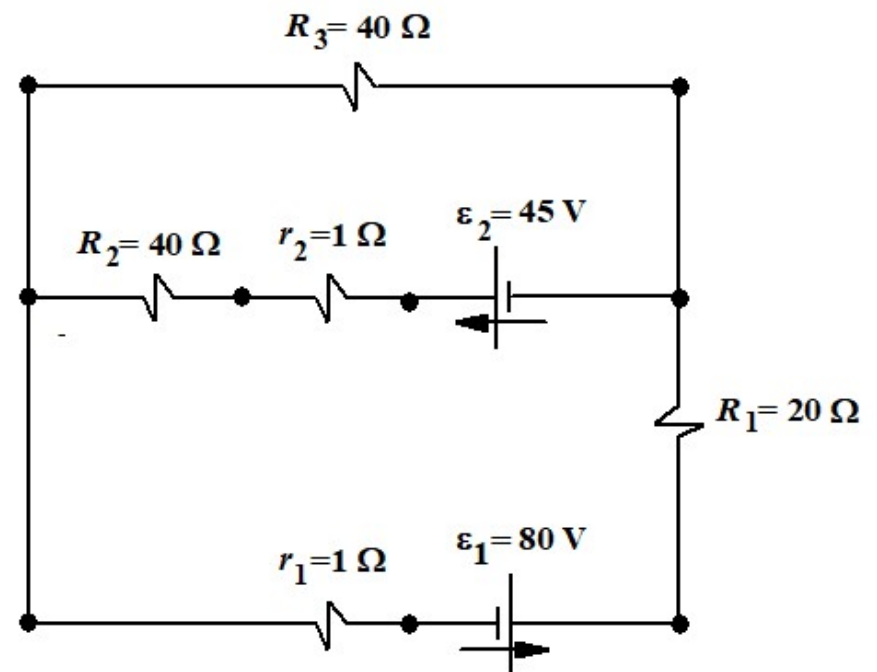
$$i_1 = 2.48A \quad i_2 = 1.78A \quad i_3 = -0.7A$$

- Then, i_3 is in the opposite direction



Example 2:

- In the shown circuit, determine the currents in all resistors using Mesh (Loop) Analysis.



Solution:

- Assume mesh (loop) currents in the same direction in both upper and lower meshes

